

## 1 Introduction

---

### 1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the **IoT-Integrated Smart Warehouse & Inventory Logistics Portal**. The document is prepared as part of the CS251 Software Engineering 1 course project (Phase 1) and serves as the primary reference for system design, development, and verification activities.

### 1.2 Scope

The system is a next-generation fulfillment management ecosystem designed to digitize the physical movement of goods within a warehouse. By integrating simulated IoT sensor data (tracking temperature, shelf weight, and location), the platform optimizes storage through Zonal Algorithms, automates procurement using the Observer Pattern, and streamlines labor through Batch Picking Logic. The system ensures that the Supply Chain is never broken, from the moment a supplier delivers raw pallets to the final second a package is scanned onto a delivery truck.

**Reference examples:** linnworks.com and shiphero.com.

### 1.3 Definitions, Acronyms, and Abbreviations

| Term   | Definition                                                       |
|--------|------------------------------------------------------------------|
| SKU    | Stock Keeping Unit – a unique identifier for each product.       |
| FEFO   | First-Expiry, First-Out – picking strategy for perishable goods. |
| HAZMAT | Hazardous Material.                                              |
| PO     | Purchase Order.                                                  |
| RBAC   | Role-Based Access Control.                                       |
| IoT    | Internet of Things – simulated sensor network in this system.    |
| SRS    | Software Requirements Specification.                             |
| FR     | Functional Requirement.                                          |
| NFR    | Non-Functional Requirement.                                      |
| DC     | Design / Implementation Constraint.                              |
| UC     | Use Case.                                                        |
| QC     | Quality Control.                                                 |
| B2B    | Business-to-Business.                                            |

### 1.4 Overview

The remainder of this document is organized as follows: Section 2 provides an overall system description. Section 3 lists all functional requirements (FR-01 – FR-42). Section 4 covers non-functional requirements. Section 5 states design and implementation constraints. Section 6 presents the Use-Case model.

## 2 Overall Description

### 2.1 Product Perspective

The portal is a standalone web-based management system for fulfillment centers. It replaces manual, paper-based warehouse processes with an integrated digital platform that connects three core operational domains: inventory storage, supplier procurement, and order fulfillment.

### 2.2 Product Functions (Summary)

The system provides 42 design functions organized into four categories:

- A. **Zonal Storage & Inventory Health** (11 functions, FR-01–FR-11)
- B. **Automated Procurement & Supplier Logic** (11 functions, FR-12–FR-22)
- C. **Picking, Packing & Routing** (10 functions, FR-23–FR-32)
- D. **System Administration & Security** (10 functions, FR-33–FR-42)

### 2.3 User Classes and Characteristics

| User Class  |                                 | Characteristics                                                                                              |
|-------------|---------------------------------|--------------------------------------------------------------------------------------------------------------|
| Warehouse   | Man-<br>ager<br><br>Floor Staff | High-level oversight of inventory health, storage efficiency, and financial procurement. Full system access. |
| Floor Staff |                                 | General warehouse personnel performing picking and packing. Restricted to operational screens only.          |
| Picker      |                                 | A Floor Staff member responsible for physically locating and collecting items from shelves.                  |
| Packer      |                                 | A Floor Staff member responsible for packing, validating, and labelling outgoing orders.                     |
| Supplier    |                                 | External B2B entity providing goods. Interacts with the procurement and invoicing modules.                   |

### 2.4 Operating Environment

- Web-based platform accessible via modern browsers.
- Handheld scanners used by Floor Staff (communicating via REST API).
- Simulated IoT sensors for shelf weight, temperature, and humidity.
- Supports multiple physical warehouse locations.

## 2.5 Assumptions and Dependencies

- All IoT sensor data is **simulated** in software; no physical hardware integration is required.
- The system assumes reliable network connectivity within the warehouse.
- Supplier interactions are modelled as B2B digital exchanges, not real third-party integrations.